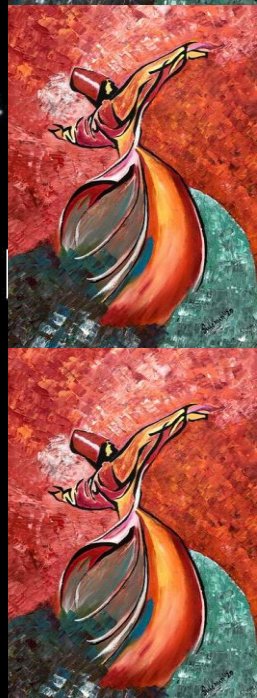




EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Spring 2026

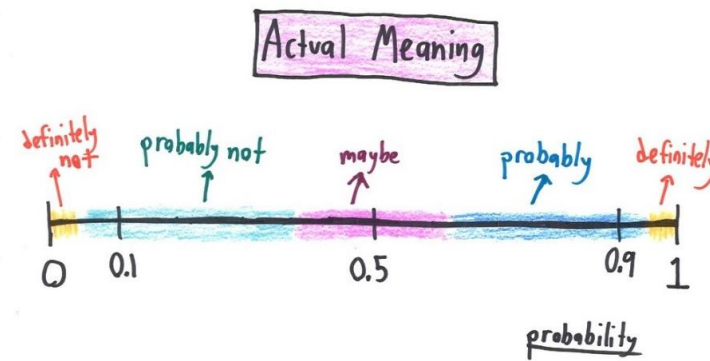




EE 354/CE 361/MATH 310 L4 section (Spring 2026)
Introduction to Probability and Statistics -

Aamir Hasan

Week 3 - Lecture No 05 – January 26, 2026



Announcements!

1. Office hours – Tuesday & Thursday 12:00 to 1:00 PM @ Room C-211 (or through meeting request)
2. Scores for Quiz No 01 posted
3. Quiz No 02 – February 04, 2026
4. Assignment No 01 to be posted by the end of week

Agenda for today

1. Unit 2 – Conditional Probability and Bayes' Rule



Meet Your TAs Rayyan Shah

About Me

I am a Computer Science Junior. I am interested in bioinformatics, AI, and game development. Outside of academics, I also enjoy cooking and baking, and I always have some new recipe to recommend. You can usually find me in the library or tapal.

Ehsaas Hours

TBD

Contact Me

✉ rso8770@st.habib.edu.pk
Or "Rayyan Ali Shah" on Teams



Meet Your TAs Hiba Aiman Ali

About Me

I am a Computer Engineering Junior with a minor in South Asian Music. My interests include bioinformatics, IoT, music and sustainability. I also sing as a part of the Habib University Orchestra. When I'm not in class, I'm most likely in the music room, singing or *trying to* learn an instrument.

Ehsaas Hours

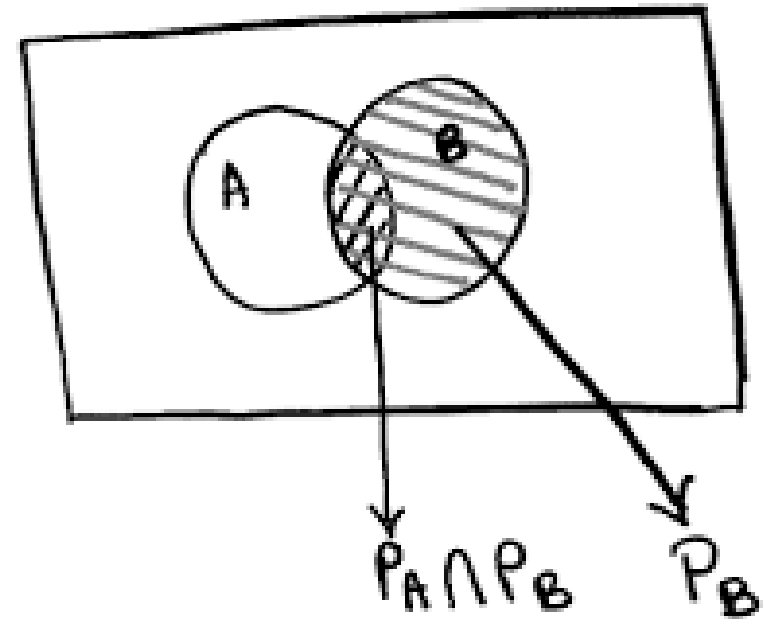
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Contact Me

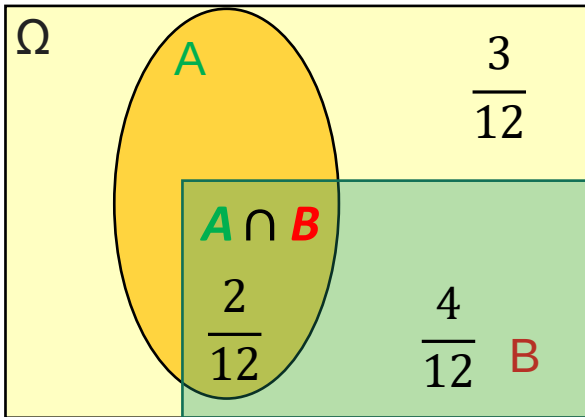
✉ ha09269@st.habib.edu.pk
on Teams or Outlook

Unit 2: Conditional Probability and Bayes' Rule

- Conditional Probability
- Three important Tools
 - Multiplication rule
 - Total probability theorem
 - Bayes' rule ($\rightarrow$ Inference)



Definition of conditional probability (Recap)



- $P(A|B)$ = “probability of A , given that B occurred”

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{2/12}{4/12} = \frac{1}{2}$$

defined only when $P(B) > 0$

Example: Models based on conditional probabilities.

- Event **A**: Airplane is flying above.
- Event **B**: Something registers on radar screen

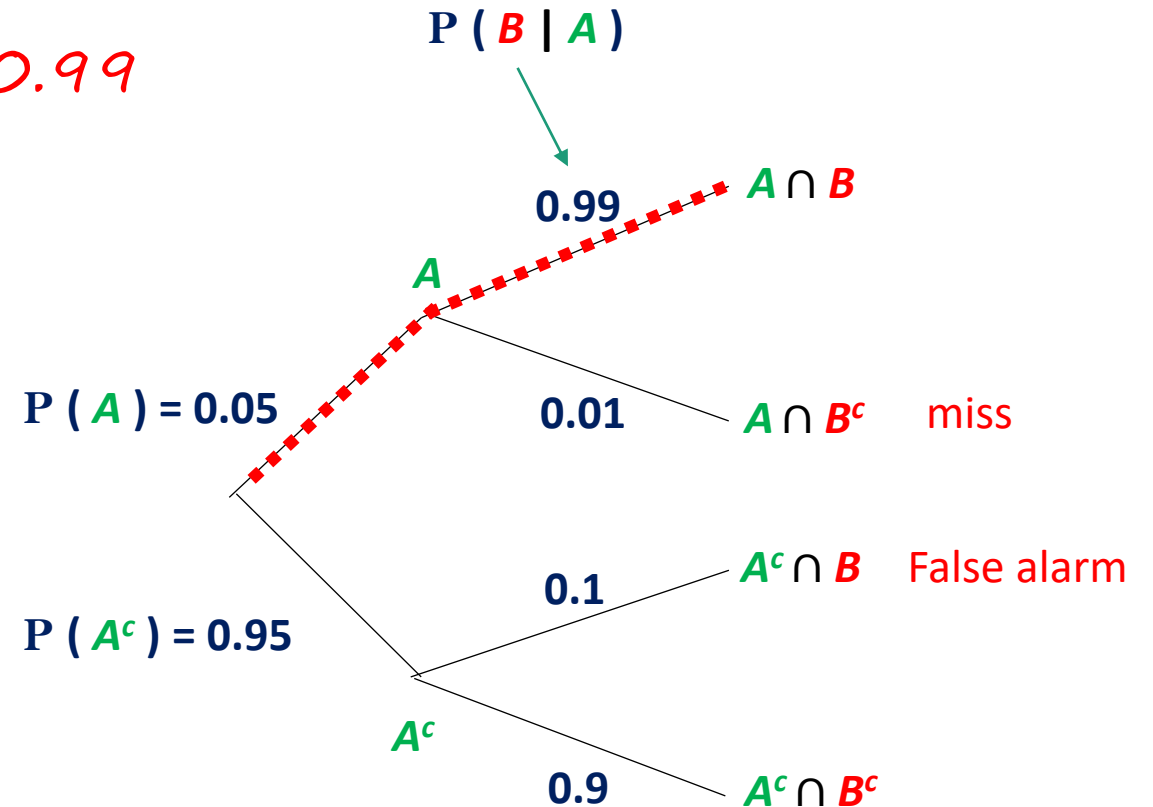
❖ Problem - Find The probability an aircraft is flying given radar registers something on radar screen – $P(A|B)$?

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$
$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$\rightarrow \bullet P(A \cap B) = P(A) * P(B|A) = 0.05 * 0.99$$

$$\rightarrow \bullet P(B) = 0.05 * 0.99 + 0.95 * 0.1 = 0.1445$$

$$\rightarrow \bullet P(A|B) = \frac{0.05 * 0.99}{0.1445} = 0.34$$



Multiplication Rule

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$\begin{aligned} P(A \cap B) &= P(A | B) P(B) \\ &= P(B | A) P(A) \end{aligned}$$

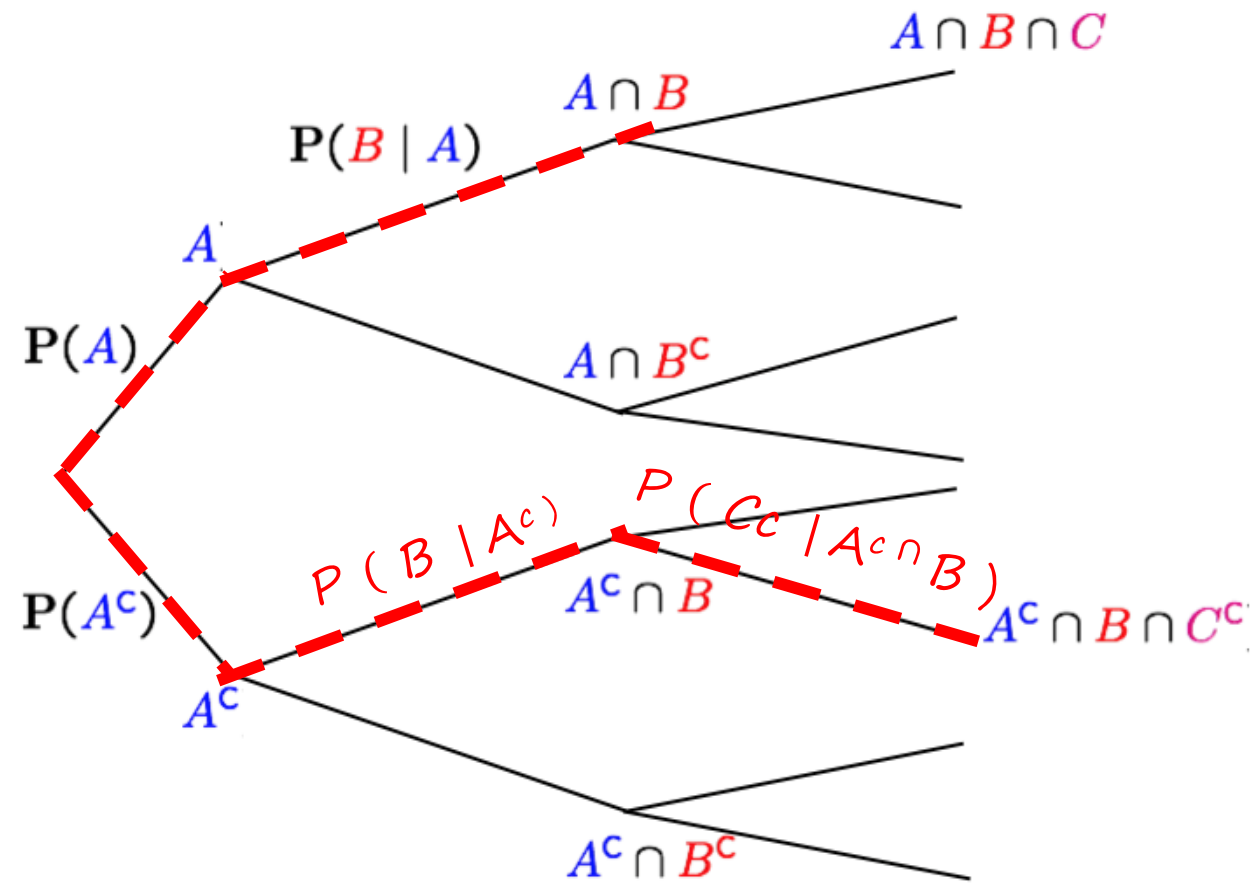
$$P(A^c \cap B \cap C^c) =$$

$$\rightarrow = P(A^c \cap B) P(C^c | A^c \cap B)$$

$$\rightarrow = P(A^c) P(B | A^c) P(C^c | A^c \cap B)$$

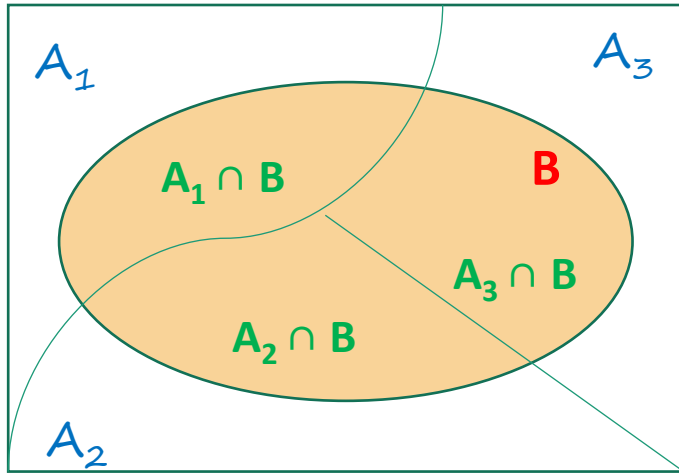
$$\rightarrow P(A_1 \cap A_2 \cap A_3 \cap \dots \cap A_n)$$

$$\rightarrow = P(A_1) \prod_{i=2}^n P(A_i | (A_1 \cap A_2 \cap \dots \cap A_{i-1}))$$



Total Probability Theorem

Ω

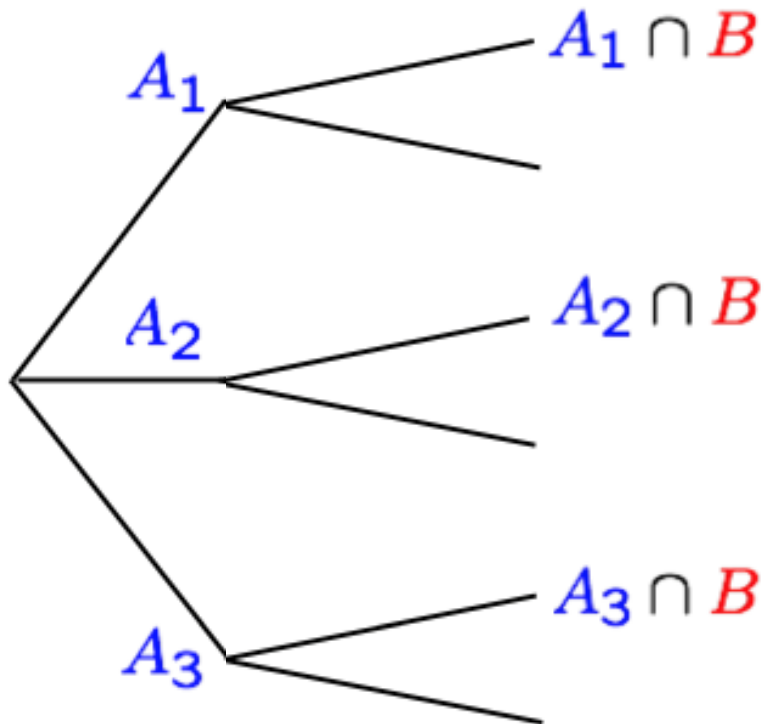


- Partition of sample space in A_1, A_2, A_3
- Have $P(A_i)$, for all i .
- Have $P(B | A_i)$, for all i .

$$\Rightarrow P(B) = P(B \cap A_1) + P(B \cap A_2) + P(B \cap A_3)$$

$$\Rightarrow = P(A_1)P(B|A_1) + \dots + \dots$$

$$\Rightarrow \sum_j P(A_j) = 1$$



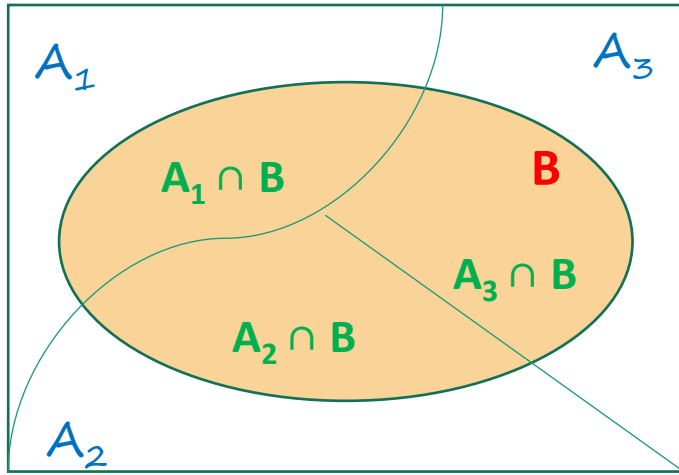
$$P(B) = \sum_j P(A_j) P(B | A_j)$$

weights

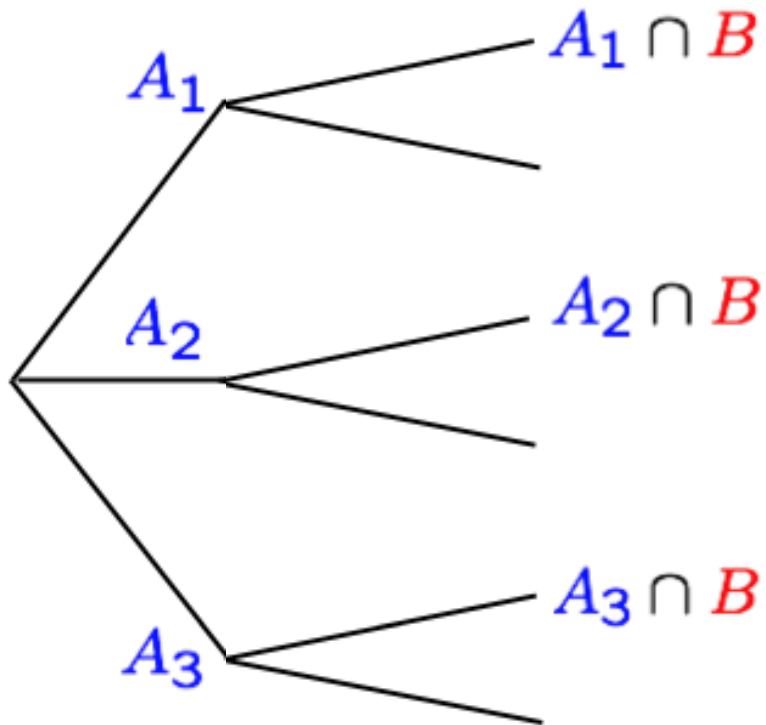
weighted average
of $P(B | A_i)$

Bayes Rule

Ω



- Partition of sample space in A_1, A_2, A_3
- Have $P(A_i)$, for all i . initial “beliefs”,
- Have $P(B | A_i)$, for all i .



$$P(A_i | B) = \frac{P(A_i \cap B)}{P(B)}$$

Revised “beliefs”, given that B occurred

$$P(A_i | B) = \frac{P(A_i) P(B | A_i)}{\sum_j P(A_j) P(B | A_j)}$$

Example: two rolls of a fair 6-sided die – conditioning Example

Y = 2 nd Roll	6						6,6
	5						
	4						
	3						
	2						
	1	1,1				5,1	
		1	2	3	4	5	6
		X = 1 st Roll					

D = two rolls resulting in doubles – (1,1), (2,2),

S = at least one of the rolls is a six

K = the rolls are of different numbers

(a) $P(D) = 6/36$

(b) $P(D | \text{Sum} \leq 4) = \frac{2/36}{6/36} = 2/6 = 1/3$

(c) $P(S) = 11/36$

(d) $P(S | K) = 10/30 = 1/3$

Example: Conditioning Example

We toss a biased coin twice with,

Event A = 1st toss is a head, Event B = 2nd toss is a head

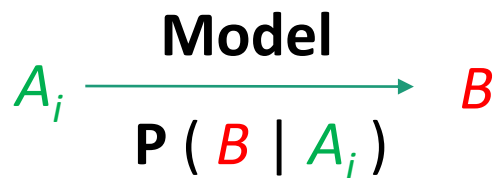
Show that $P(A \cap B \mid A) \geq P(A \cap B \mid (A \cup B))$

Bayes' Rule & inference

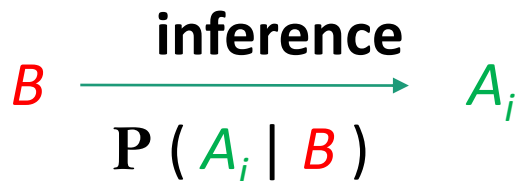
- ❖ Thomas Bayes, Presbyterian minister (c. 1701-1765)
- ❖ “Bayes’ Theorem” published posthumously
- ❖ Systematic approach for incorporating new evidence

➤ Bayesian Inference

- Initial beliefs $P (A_i)$ on all causes of an observed event B .
- Model of the world under each A_i : $P (B | A_i)$



- Draw conclusions about causes



$$1) P (A | B) = \frac{P (A \cap B)}{P (B)}$$

$$2) P (B | A) = \frac{P (A \cap B)}{P (A)}$$

$$3) P (B | A) = \frac{P (A | B) \cdot P (B)}{P (A)}$$

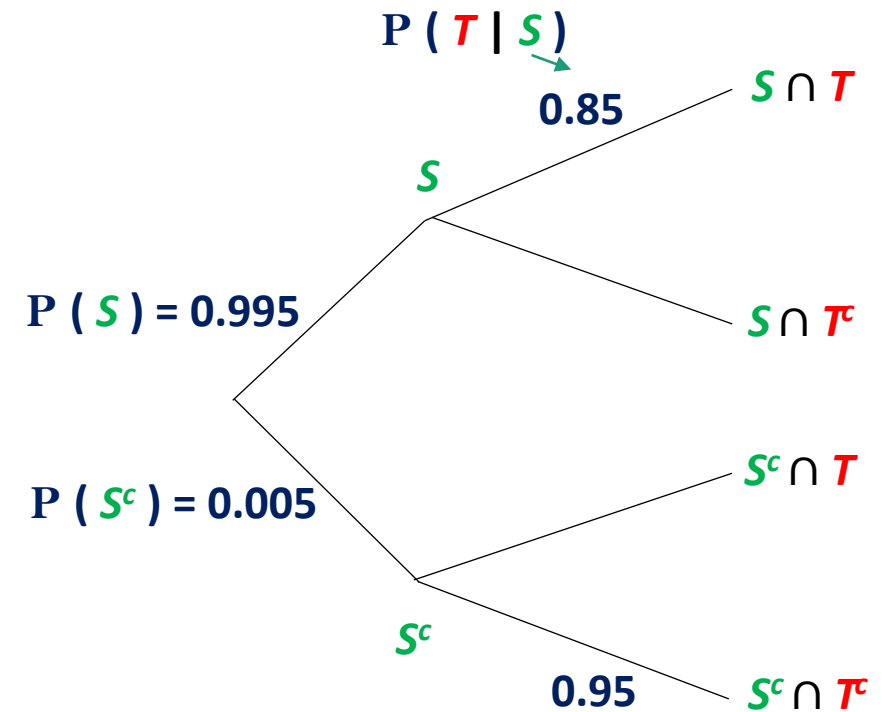
Example: Baye's Rule

A new test (T) has been developed to determine whether a given student is overstressed (S). This test is 95% accurate if the student is not overstressed, but only 85% accurate if the student is in fact overstressed. It is known that 99.5% of all students are overstressed. *Given that a particular student tests negative for overstress, what is the probability that the test results are correct, and that this student is not overstressed?*

S = Student is over-stressed, S^c = Student is NOT over-stressed

T = Test positive, T^c = Test negative

$$\begin{aligned}\text{Find } P(S^c | T^c) &= \frac{P(T^c | S^c) \times P(S^c)}{P(T^c)} \\ &= \frac{0.95 \times 0.005}{0.95 \times 0.005 + 0.15 \times 0.995} \\ &= 0.0308\end{aligned}$$



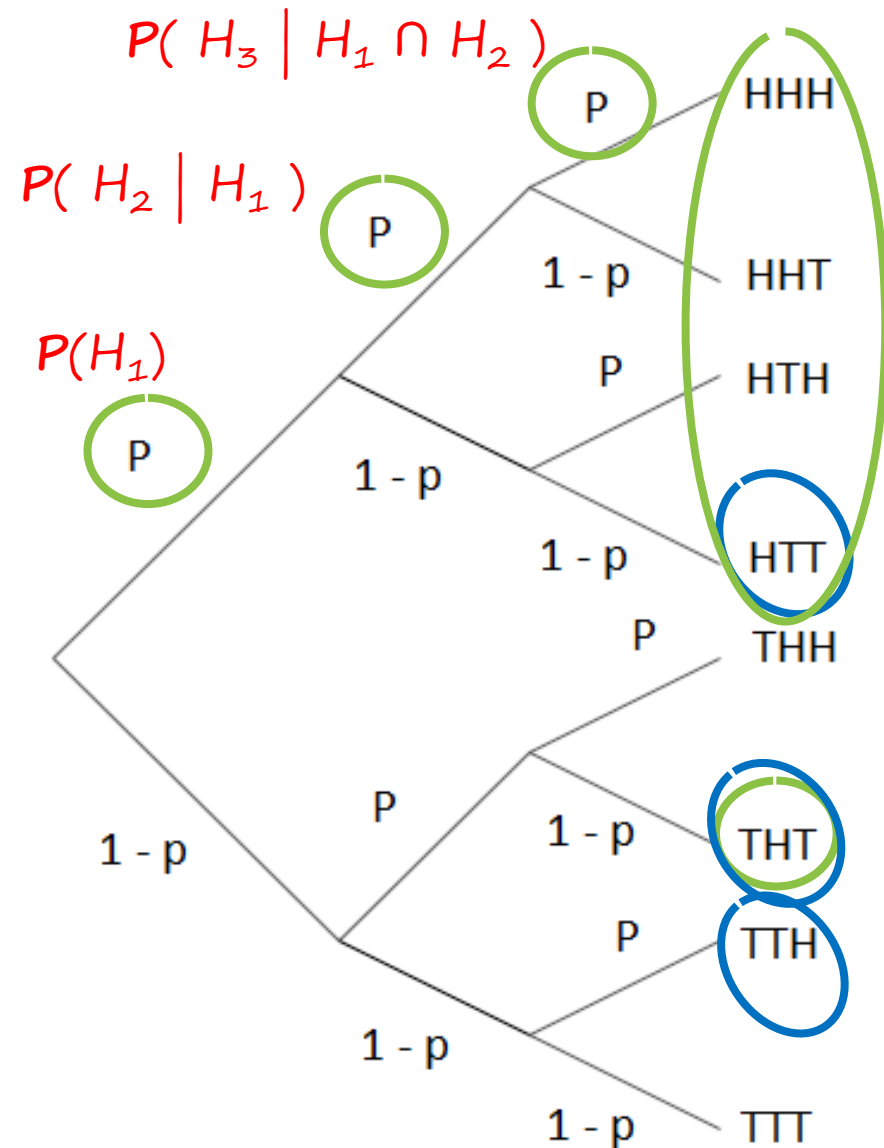
1) $P(A|B) = \frac{P(A \cap B)}{P(B)}$

2) $P(B|A) = \frac{P(A \cap B)}{P(A)}$

3) $P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A)}$

A model based on conditional Probabilities

Let's consider 3 tosses of biased coin: $P(H) = p$; $P(T) = 1 - p$



1. Multiplication Rule: $P(\underline{THT}) = (1-p) p (1-p)$

2. Total Probability: $P(1 \text{ Head}) = 3 p (1-p)^2$

3. Bayes Rule:

$$\begin{aligned} P(\underline{H_1} | 1 \text{ Head}) &= \frac{P(H_1 \cap 1 \text{ head})}{P(1 \text{ head})} \\ &= \frac{p(1-p)^2}{3 p(1-p)^2} = \frac{1}{3} \end{aligned}$$

Few Comments

$$P(H_2 | H_1) = p = P(H_2 | T_1)$$

Any deductions?

$$\begin{aligned} P(H_2) &= P(H_1) P(H_2 | H_1) + P(T_1) P(H_2 | T_1) \\ &= p \cdot p + (1-p) \cdot p \\ &= p \end{aligned}$$

$$\text{So, } P(H_2 | H_1) = p = P(H_2 | T_1)$$